IN3240/IN4240 Software Testing

Tool support for testing

Chapter 6

1. Test tool considerations and classification
2. Potential benefits and risks
3. Effective use of tools – introducing a test tool to an organization



1. Types of test tools
- *1.1 Tool support for testing*
- *1.2 Test tool classification*
- *1.3 Tools for test management*
- *1.4 Tools for static testing*
- *1.5 Tools for test design and specification*
- *1.6 Tools for execution and logging*
- *1.7 Tools for performance & monitoring*
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2. Effective use of test tools
- *2.1 Potential benefits and risks of tools*
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Potential benefits and risks

- Simply purchasing or leasing a tool does not guarantee success with that tool!
- Each type of tool may require additional effort to achieve real and lasting benefits.



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Potential benefits and risks

Greater consistency and repeatability

People tend to do the same tasks in a slightly different way

Distractions affect human performance

Doing more than one task simultaneously

Interruptions by peers / co-workers

Fatigue and personals issues

External pressures

Tools will reproduce the exact same procedure as previously



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Potential benefits and risks

Objective assessment

Humans are prone to make errors

Subjective preconceived notions and bias toward verification

Testing tools on the other hand …

Objective "preconceived notions"

Assessment → Repeatable and consistently calculated

Cyclomatic complexity, nesting levels

Coverage, system behaviour, incident statistics



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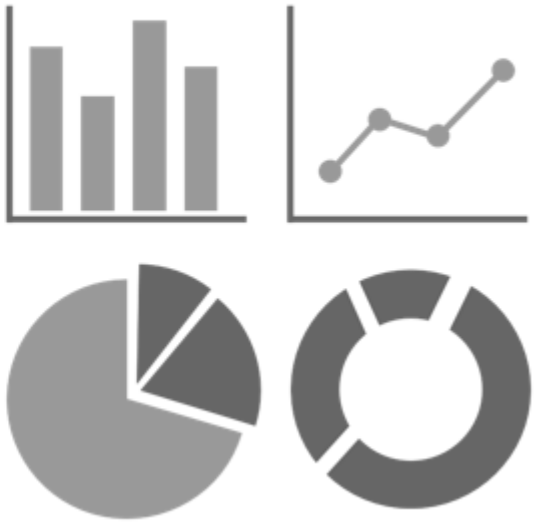
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Potential benefits and risks

Ease of access to information about the tests or test effort

- Information presented visually
- Easier for the human mind to understand
  - Chart, graphs > Long list of numbers
- Special purpose tools provide features directly
  - Statistics and graphs
  - Incident rates
  - Performance



Potential benefits and risks

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Potential benefits of using tools :

- Reduced repetitive work (running regression tests, re-entering the same test data. Etc)
- Greater consistency and repeatability (tests executed by a tool, tests derived from requirements).
- Objective assessment (static measures, coverage).
- Ease of access to information about tests or testing (statistics / graphs about test progress, incident rates, performance)

Potential benefits and risks

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2. Effective use of test tools

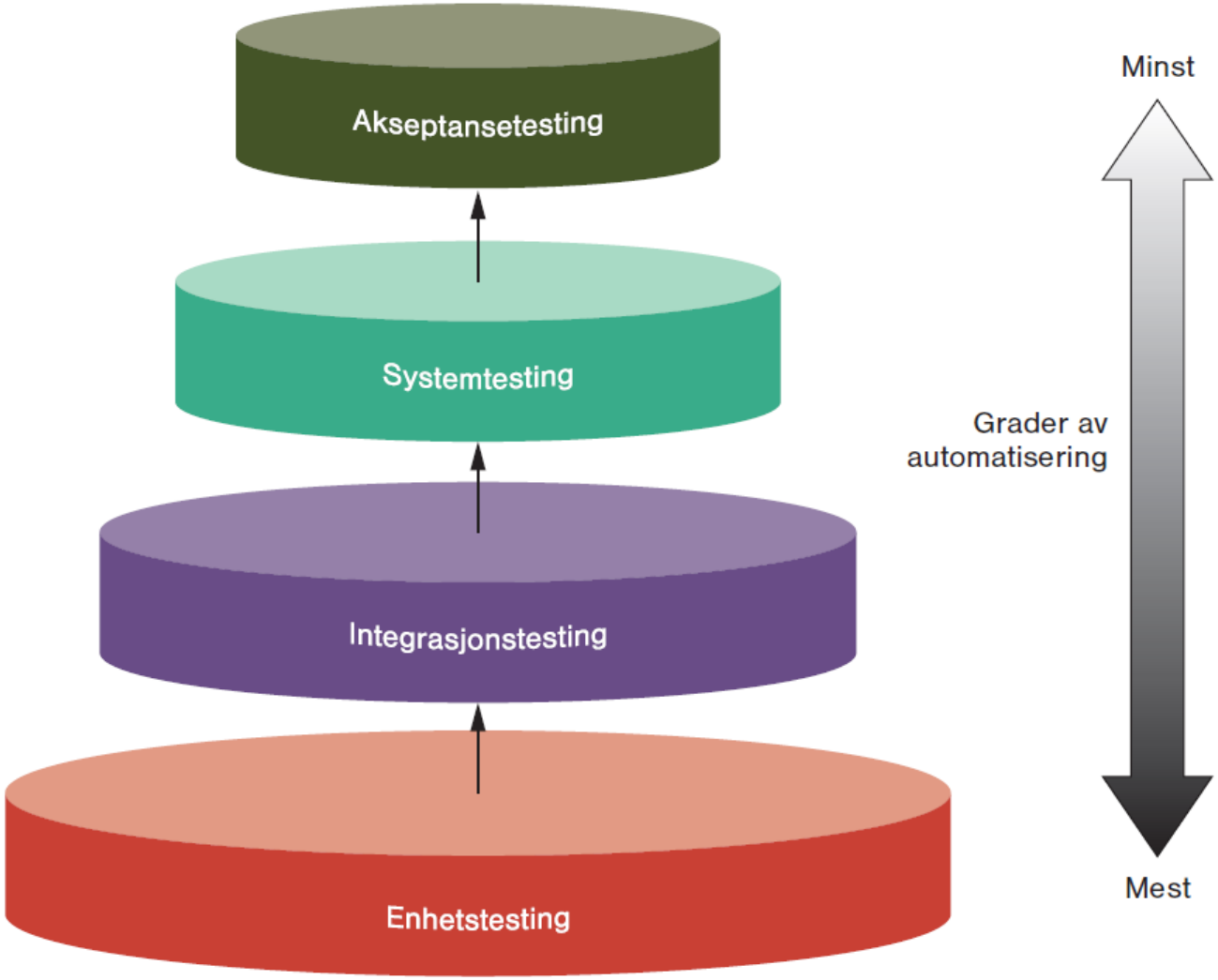
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Potential risks of using tools:

- Unrealistic expectations for the tool (functionality & ease of use).
- Underestimating time, cost and effort for the introduction of a tool (training, external expertise).
- Underestimating the time and effort needed to achieve significant and continuing benefits from the tool
- Underestimating the effort required to maintain the test assets generated by the tool.
- Over-reliance on the tool (replacement where manual testing would be better).

Test automation



Figur 16.1: Anbefalt testautomatiseringsgrad.

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Potential benefits and risks

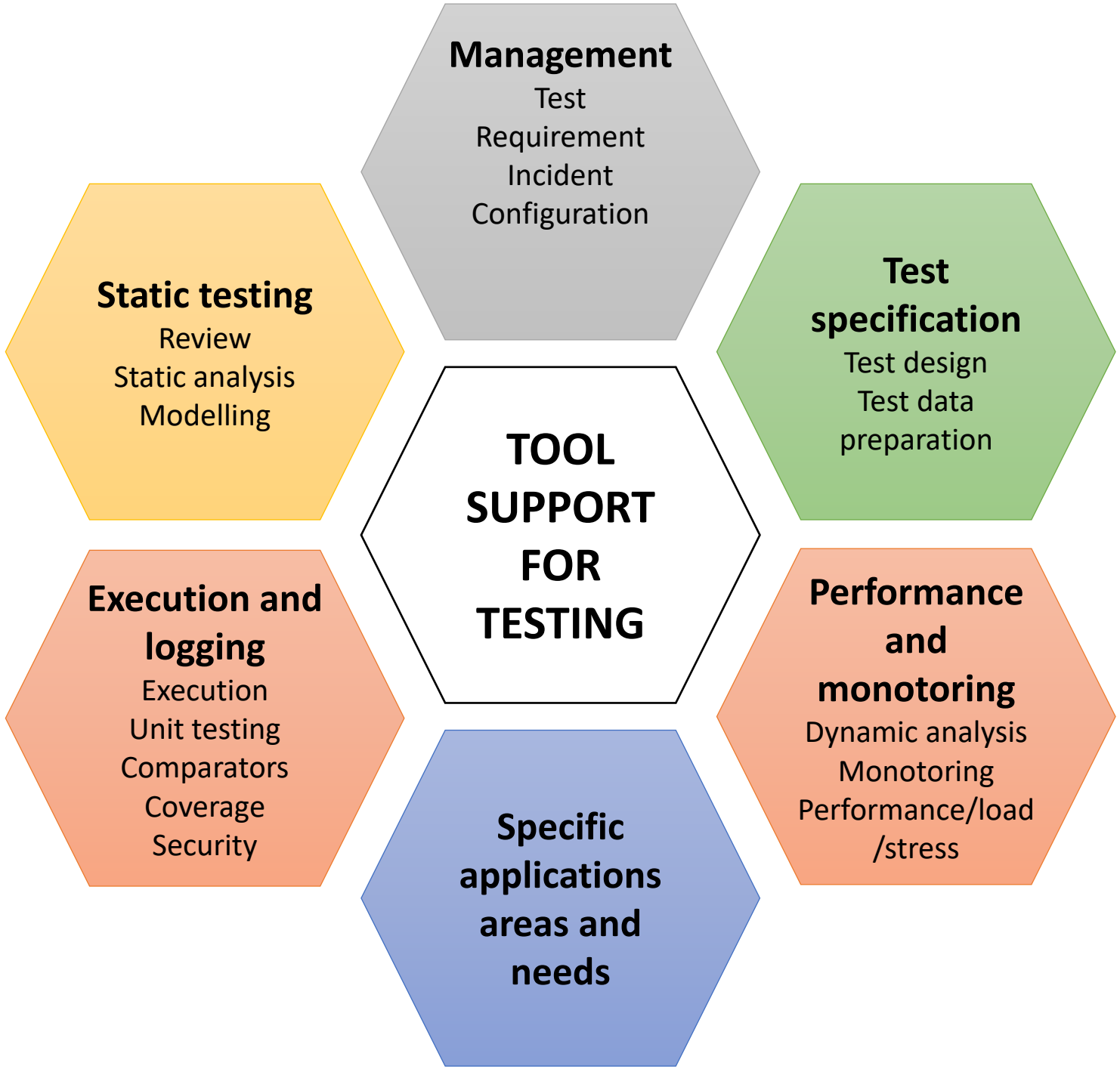
Tools are not magic!

They can do very well what they have been designed to do, but they cannot do everything.

The tester concentrates on
- what should be tested
- what the test cases should be
- how to prioritize the testing

The tool user concentrates on
- how best to get the tool to do its job effectively
- how to give increasing benefit from tool use




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Tool support for testing – types of tools

Test tools can be used for one or more activities that support testing:

- Tools that are directly used in testing (e.g. test execution tools, test data generation tools, result comparison tools)
- Tools that help in managing the testing process (e.g. test results, requirements, incidents, defects) and for monitoring and reporting the test execution
- Tools that are used in exploration (e.g. tools that monitor the file activity for an application)
- Any tool that aids in testing

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Tool support for testing - purposes

- **Tools support** for testing can have one or more of the following purposes, depending on the context:
  - improve the efficiency of the test activities (e.g.: by automating repetitive tasks)
  - automate activities that require significant resources when done manually (e.g. static testing)
  - automate activities that cannot be done manually (e.g. large-scale performance testing of client-server applications)
  - increase reliability of testing (by automating large data comparisons or simulating complex behavior)



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Test tool classification

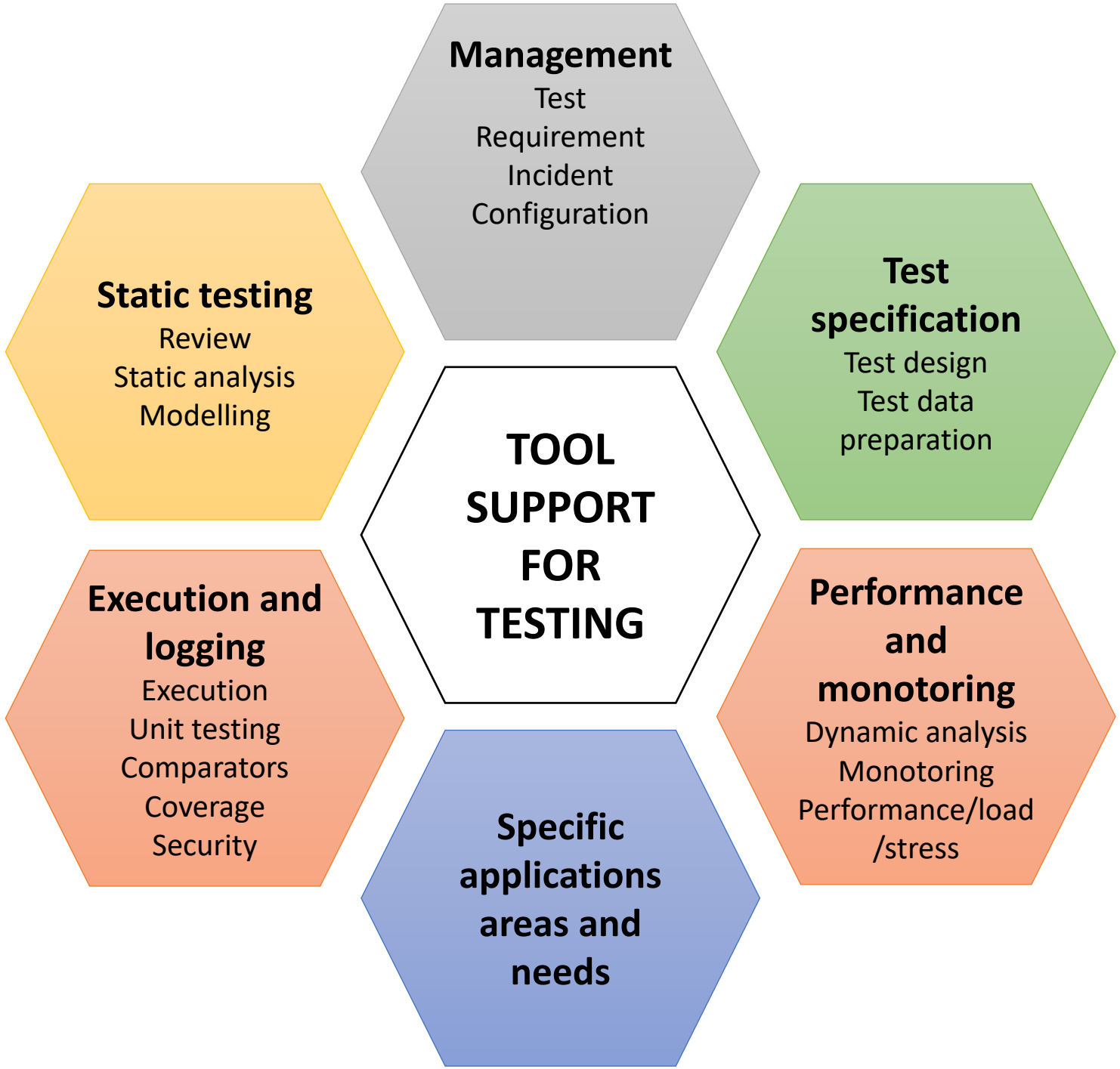
Tools are classified according to the testing activities that they support.

- one activity
- more than one activity, but classification falls under the main activity

Notes

- Some types of test tool can be intrusive - the tool itself can affect the outcome of the test. (e.g. timing measurements may be different depending on how you measure it with different performance tools).
- The consequence of intrusive tools is called the probe effect.
- Some tools offer support more appropriate for developers. Such tools are marked with "(D)" in this chapter.




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Tools support for management of testing & tests

Characteristics

- Support for the management of tests and the testing activities.
- Support for traceability of tests, test results and incidents to source documents, such as requirements specifications.
- Generation of progress reports
- Logging test results
- Monotoring
- Offer info on metrics related to the tests.



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Tools support for management of testing & tests

Types

Test management tools and
application lifecycle management tool (ALM)

- Requirements management tools
- Defect management tools
- Configuration management tools
- Continuous integrations tool (D)



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Tools support for management of testing & tests

Types

- Test management tools and application lifecycle management tool (ALM)

Requirements management tools

- Store requirements
- Check for consistency and undefined (missing) requirements
- Allow prioritization
- Enable individual tests to be traceable to requirements

- Defect management tools

- Configuration management tools

- Continuous integrations tool (D)



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- Statuses of incident reports:

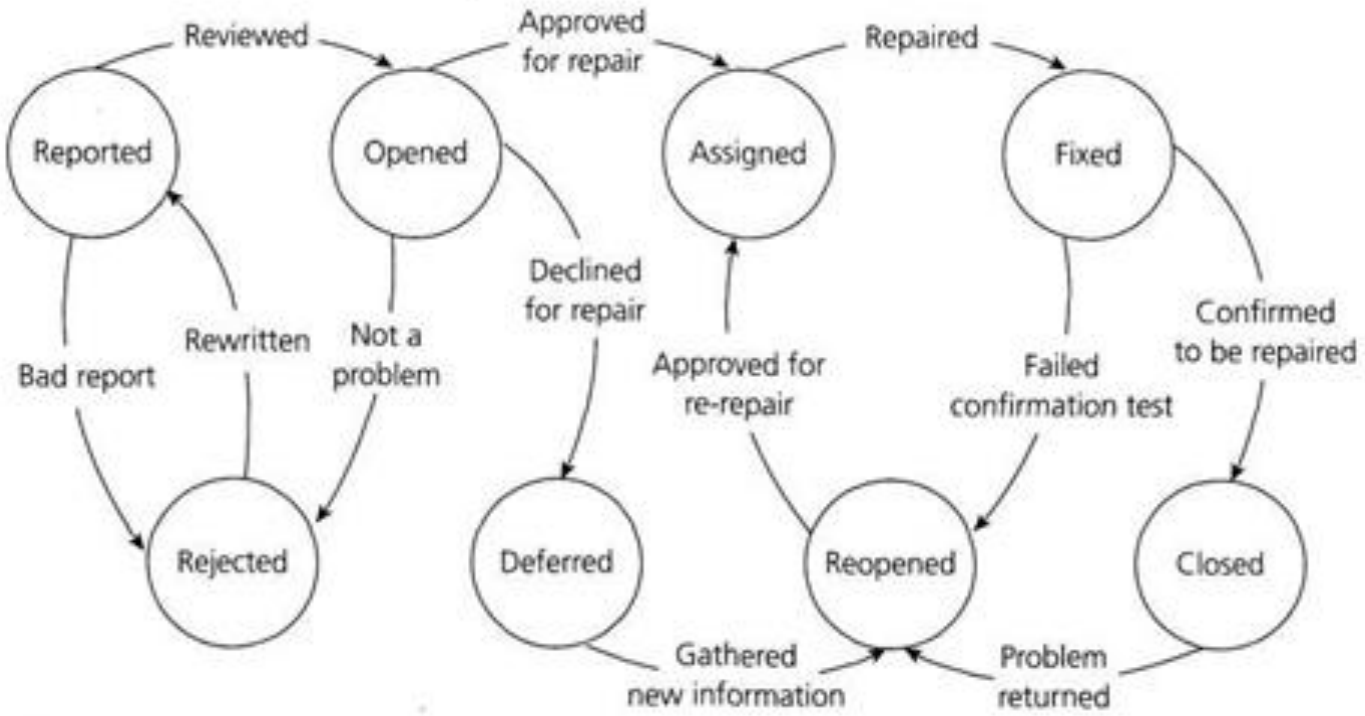


FIGURE 5.3 Incident report life cycle

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Tools support for management of testing & tests

Types

- Test management tools and application lifecycle management tool (ALM)
- Requirements management tools
- Defect management tools

Configuration management tools

- Are necessary to keep track of different versions and builds of SW and tests
- Are particularly useful when developing on more than one configuration of the HW/SW environment

- Continuous integrations tool (D)



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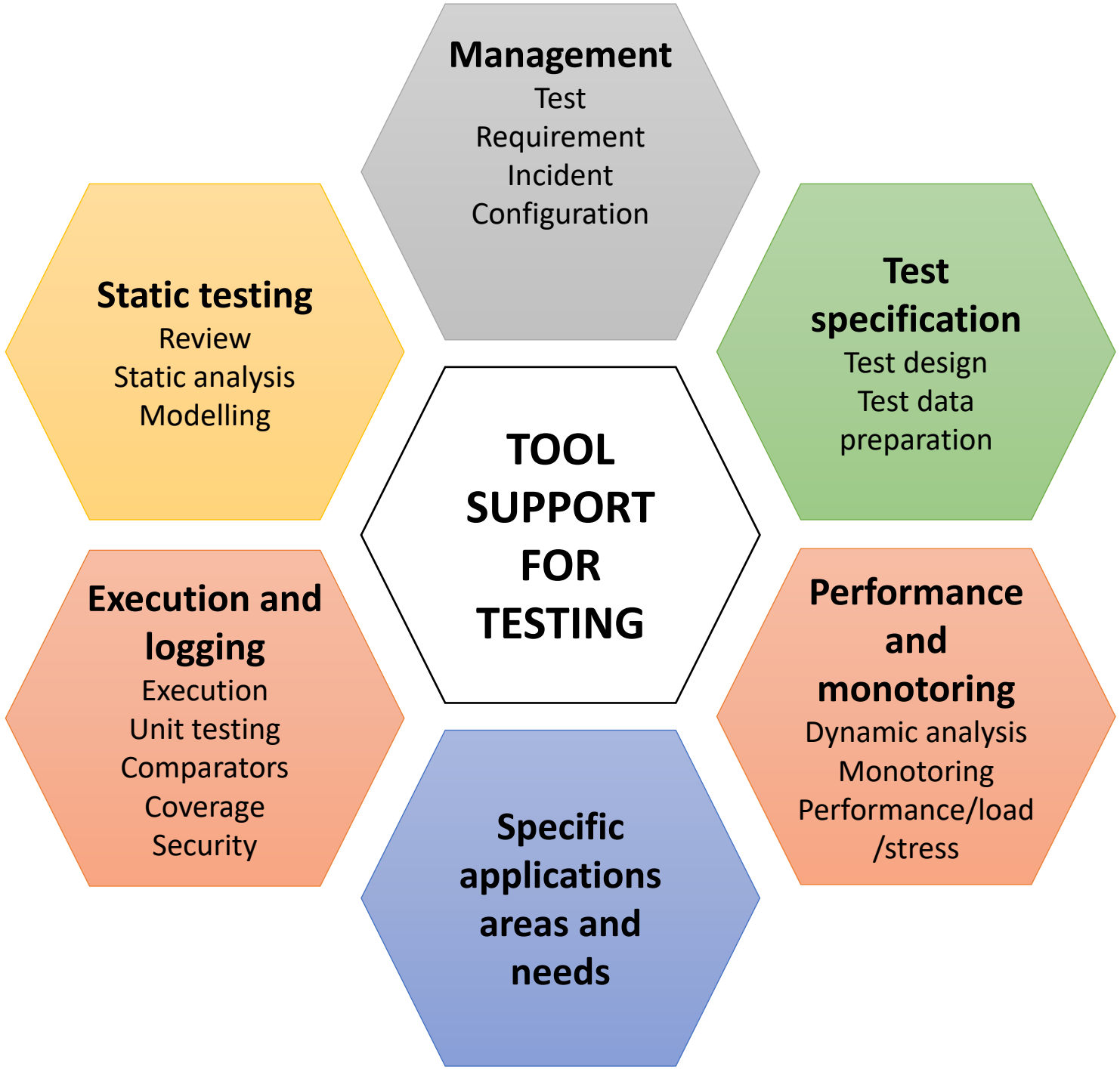
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Continuous integrations tool (D)


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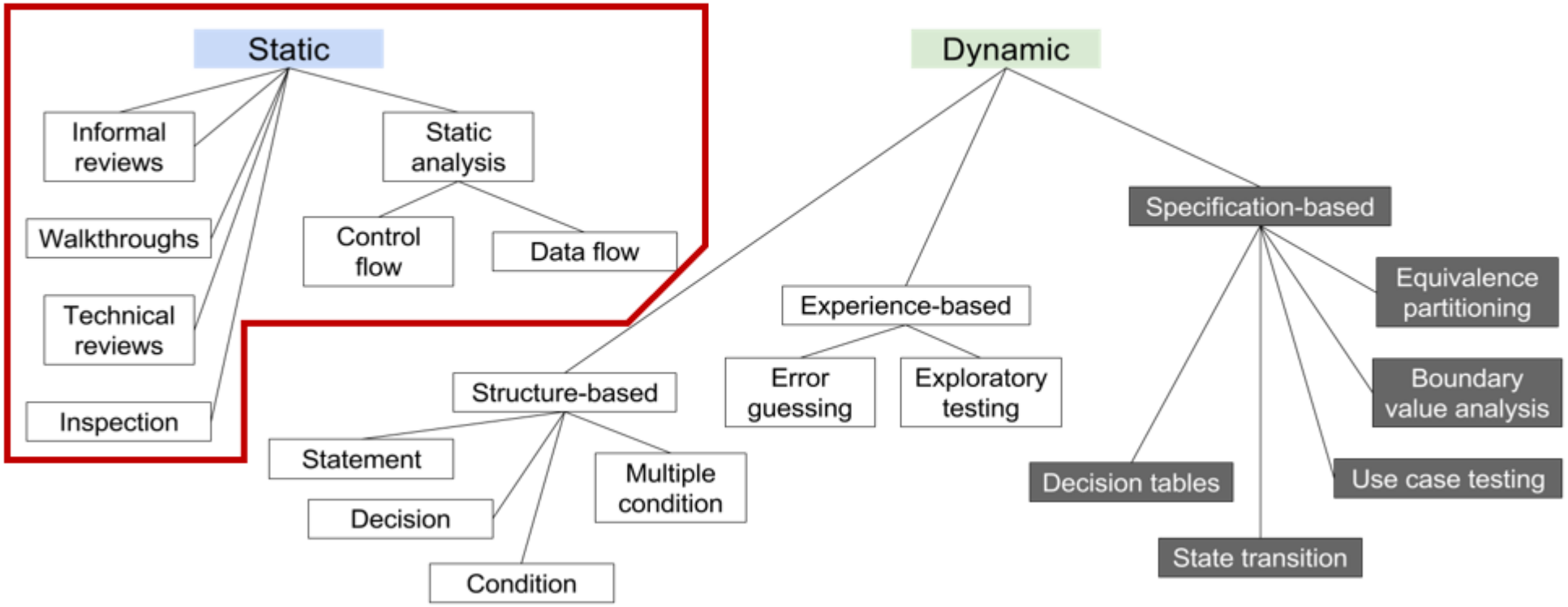
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Tools support for static testing

Recall the different types of testing



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Tools support for static testing

Tools for static testing

Tools that aid in improving the code / work product, without executing it

Categories

Review tools

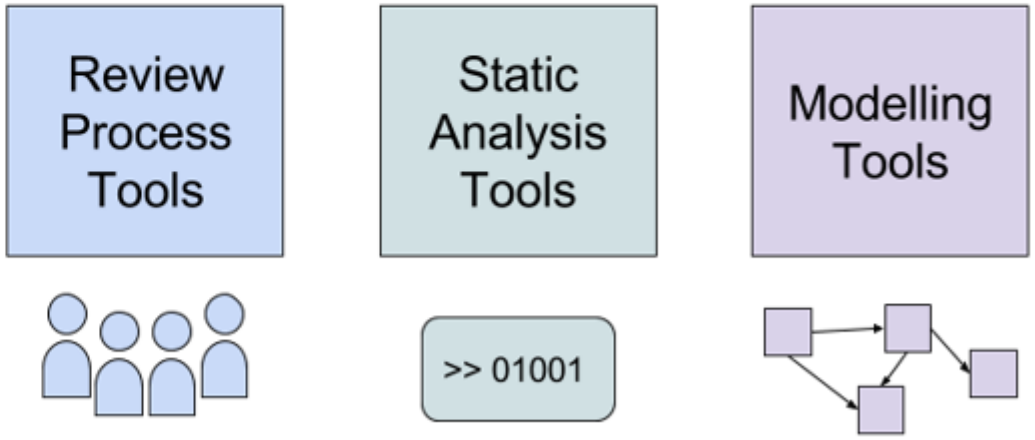
Supports the review process

Static analysis tools

Supports code examination

Modelling tools

Validate models of system / software



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Tools support for static testing



Review process tools

Common reference for the review processes conducted

Keep track of all the information from the review process

Store and communicate review comments, report on defects and effort

Monitoring review status → Passed, passed with corrections, requires re-review

When to use?

Suitable for more formal review processes

Geographically dispersed teams

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Tools support for static testing

Static analysis tools (D)

Static Analysis Tools

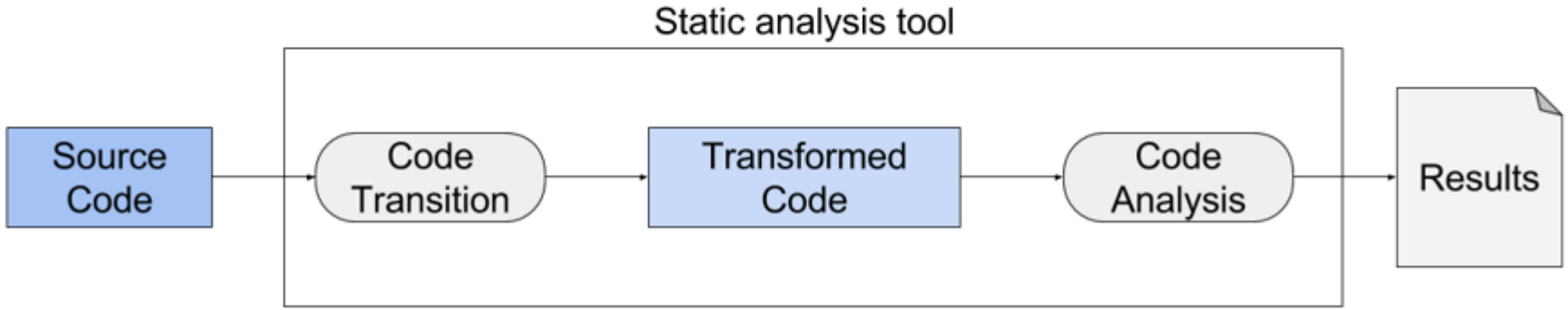
Mostly used by developers → Component (unit) testing

Tool is executed → Code is not

>> 01001

The source code serves as input data to the tool

Extension of compiler technology



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Tools support for static testing

Static analysis tools (D)

Support developers and testers in finding defects before dynamic testing

Purpose

To better understand the code, and find ways of improving it

Common features

Calculate metrics → Complexity, nesting levels → Identify areas of risk

Enforce coding standards

Analyse code structures and dependencies



Static analysis tools (D)

Static analysis tool example: Source Monitor

Collects metrics from source code files

Displays and prints metrics in tables and charts

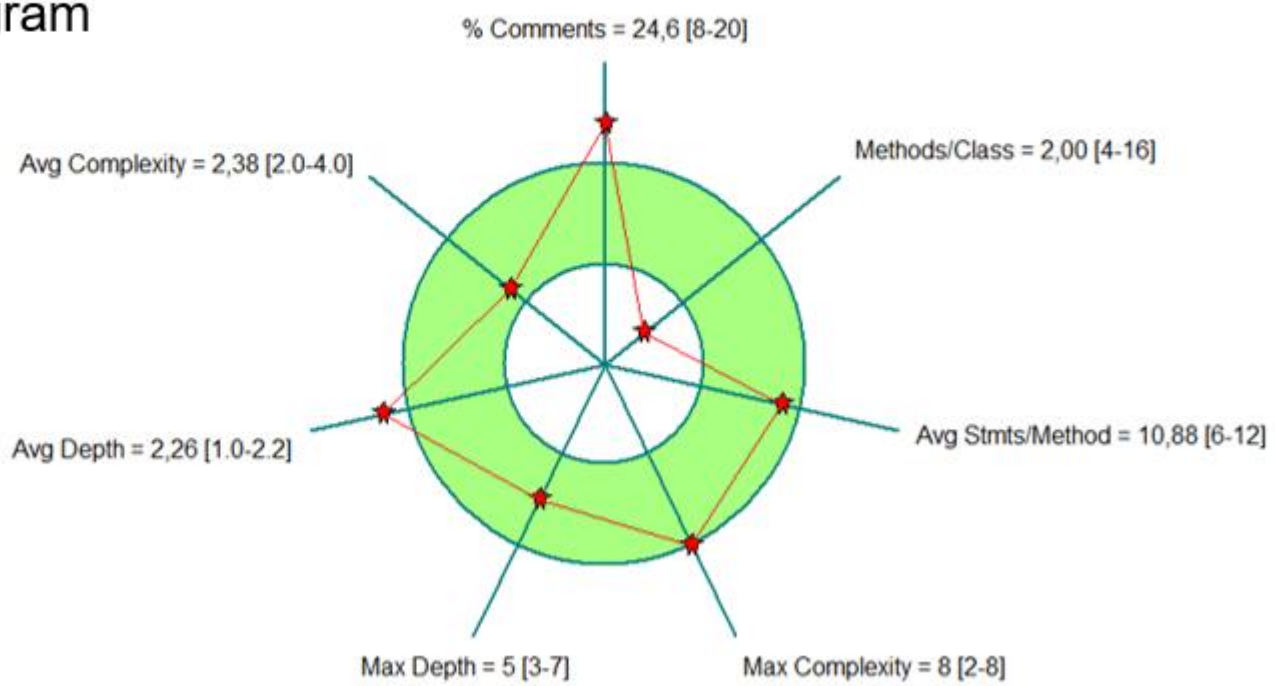
File Edit View Window Help

File Name	Li...	Statements	% Branches	Calls	% Comments	Classes	Methods/Class	Avg Stmts/Method	Max Complexity	Max Depth	Avg Depth	Avg Complexity
GUI.java	195	110	10,0	69	24,6	4	2,00	10,88	8	5	2,26	2,38
Board.java	104	46	0,0	40	36,5	1	1,00	19,00	1	2	1,00	1,00
Square.java	86	50	8,0	6	19,8	1	15,00	1,80	3	3	1,56	1,27
EmptySquare.java	70	25	44,0	17	20,0	1	3,00	7,00	7	5	2,80	4,67
UtenGUIMain.java	63	30	23,3	23	20,6	1	1,00	26,00	8	4	2,83	8,00
SquareContainer.java	32	17	5,9	1	18,8	1	4,00	2,25	2	2	1,47	1,25
SudokuContainer.java	29	11	0,0	3	27,6	1	4,00	0,75	1	2	1,09	1,00
FilledSquare.java	27	8	25,0	5	29,6	1	2,00	2,50	3	3	1,75	2,00
Column.java	13	3	0,0	1	38,5	1	1,00	1,00	1	2	1,00	1,00
Row.java	13	3	0,0	1	38,5	1	1,00	1,00	1	2	1,00	1,00
Box.java	12	3	0,0	1	41,7	1	1,00	1,00	1	2	1,00	1,00

Static analysis tools (D)

Static analysis tool example: Source Monitor

Kiviat diagram



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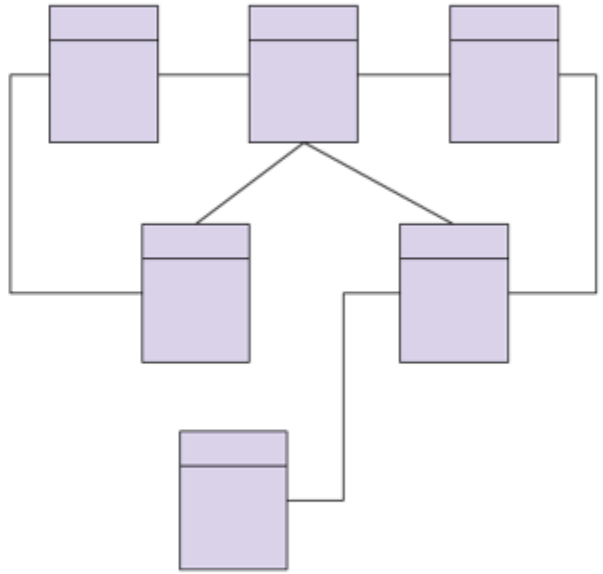
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Tools support for static testing

Modelling tools (D)

Validate models of the system / software



Purpose

To better aid in designing the software

Common features and characteristics

Identify inconsistencies and defects within the models

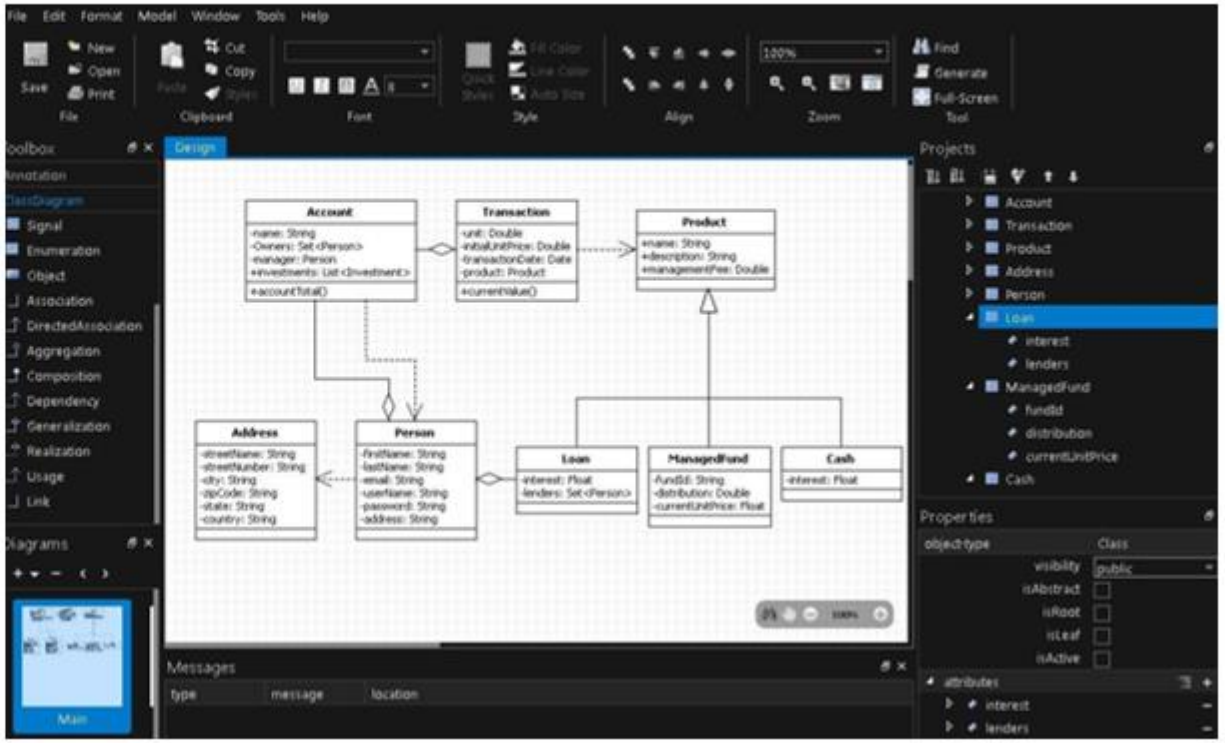
Identify and prioritise risk areas

Predicting system response and behaviour under various situations

Tools support for static testing

Modelling tool example: Star UML

- UML tool
- Variety of diagrams
  - Class / Domain
  - Use case
  - Sequence



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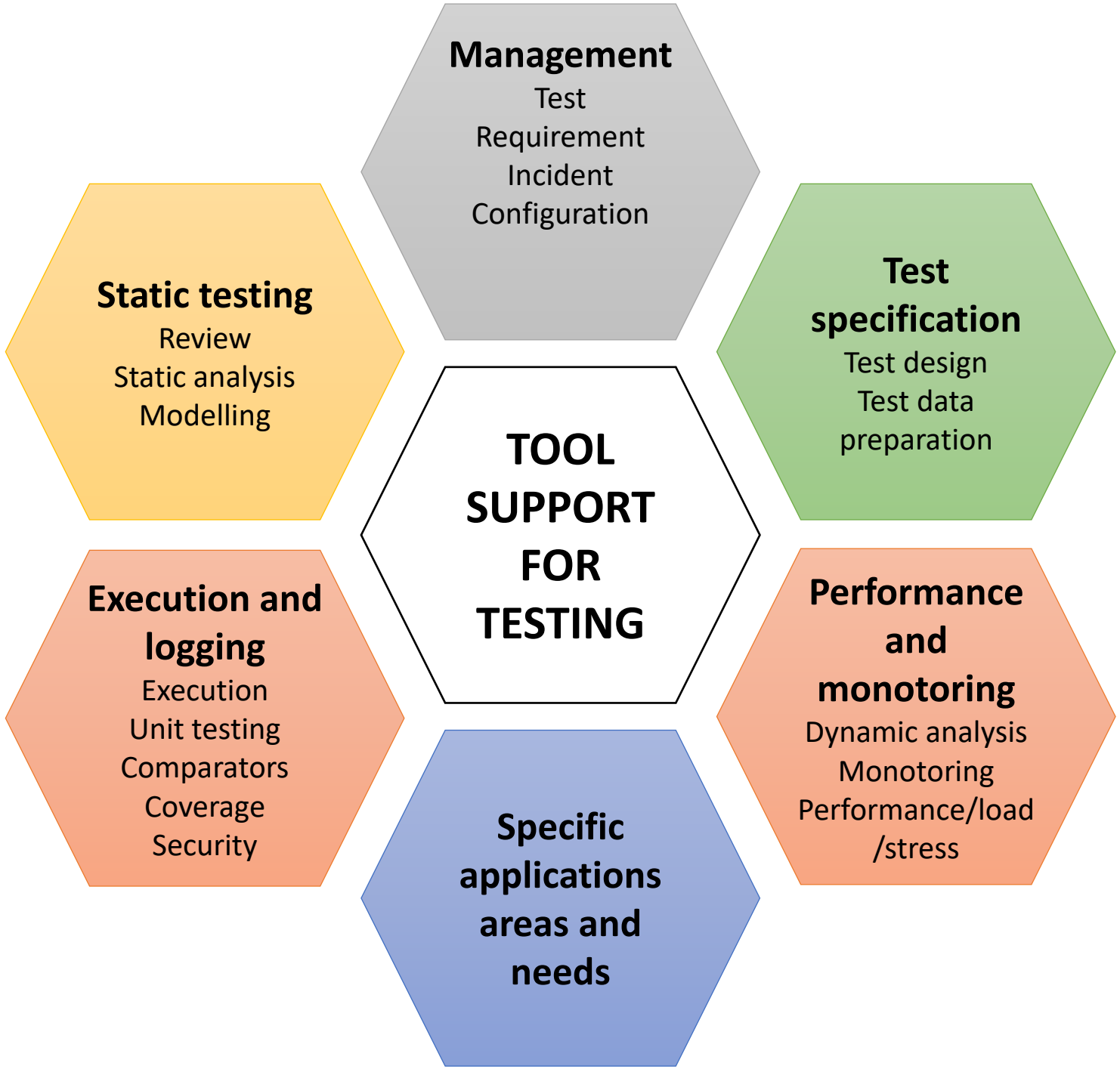
2. Effective use of test tools
- *2.1 Potential benefits and risks of tools*
- *2.2 Special considerations for tools*

3. Introducing a test tool into an organization

Tools support for static testing

The major benefit of static testing tools and modeling tools is **the cost effectiveness** of finding more defects at an **earlier time** in the development process.

As a result, the development process may accelerate and improve by having less rework.


Management
Test
Requirement
Incident
Configuration
Static testing
Review
Static analysis
Modelling
Test specification
Test design
Test data preparation
TOOL SUPPORT FOR TESTING
Execution and logging
Execution
Unit testing
Comparators
Coverage
Security
Performance and monotoring
Dynamic analysis
Monotoring
Performance/load /stress
Specific applications areas and needs

1. Types of test tools
- *1.1 Tool support for testing*
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3. Introducing a test tool into an organization

Tools support for test design and specification

- Test design tools
- Model-based testing tools
- Test data preparation tools
- TDD - Test driven development tool (D)
- ATDD - Acceptance test-driven development and
  BDD - Behavior-driven development



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3. Introducing a test tool into an organization

Tools support for test design and specification

Test design tools

- generating test inputs values from:
  - requirements
  - test conditions
  - design models (state, data or object)
  - code
  - graphical user interfaces
- generating expected results, if an oracle is available to the tool.

- Model-based testing tools
- Test data preparation tools
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Tools support for test design and specification

- Test design tools

Model-based testing

- generating test inputs values or test cases from stored information the describes a model of the system, e.g. state transition model

- Test data preparation tools

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Tools support for test design and specification

- Test design tools
- Model-based testing tools

Test data preparation tools
- Generating extensive range or volume of data, if needed
- Manipulate databases or files to set up test data to be used during the execution of the tests

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3. Introducing a test tool into an organization

Tools support for test design and specification

Which test activities are supported by test data preparation tools?

Common features of test data preparation tools

- Data can be selected from an existing database
- Data can be created, generated, and altered for use in tests
- Construct a large number of similar records → Volume tests

When to use?

- During test specification and control → Test data management is difficult
- Ensure the system under test is being tested *realistically*
- Useful for performance and reliability testing

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Tools support for test design and specification

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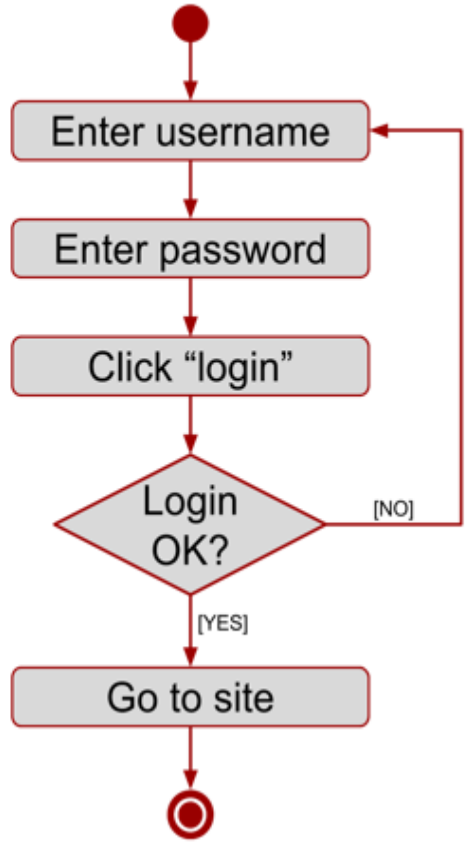
Example: Simple login site



Data need for testing website

- List of various usernames
- List of various passwords
- Database of existing users

We do not want to create all this by hand!



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Tools support for test design and specification

- Test design tools
- Model-based testing tools
- Test data preparation tools

TDD - Test driven development tool (D)
- part of EXtreme Programming
- used in Agile development
- Recall the first guest lecture!

- ATDD - Acceptance test-driven development and
  BDD - Behavior-driven development



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ATDD - Acceptance test-driven development and

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- Natural language syntax, Given/When/Then:

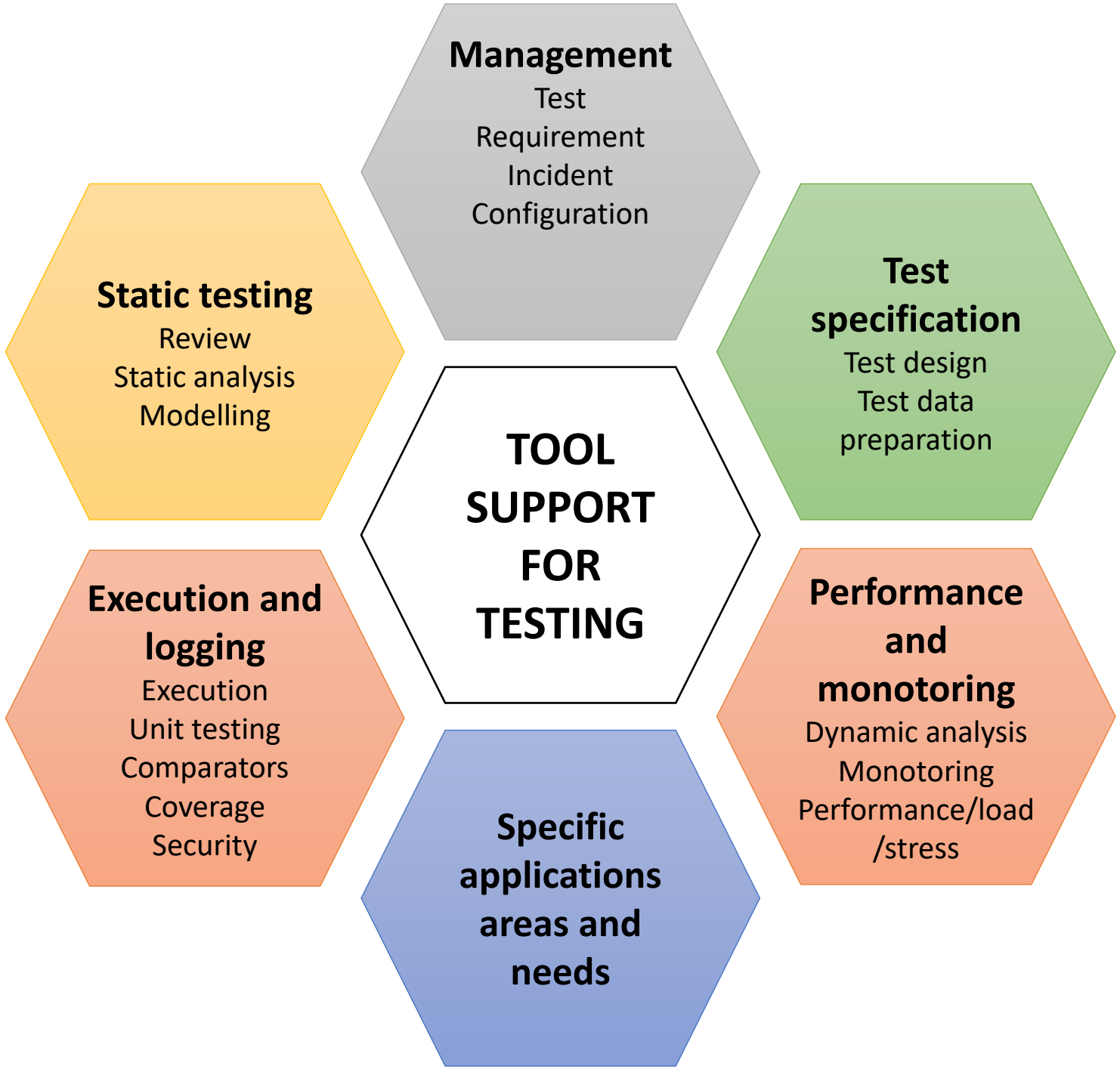
  Given <some condition> , When <something is done>, Then <result should happen>

  As a <role, e.g. costumer>

  In order to <achieve something, e.g. product>

  I want <do something>




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Tools support for test execution & logging

Test execution tools

- Enable tests to be executed automatically using stored
  - inputs and
  - expected results

- Coverage tools

- Test harnesses (D) and Unit test framework tools (D)

- Test comparators



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3. Introducing a test tool into an organization

Tools support for test execution & logging

Levels of scripting

- Linear Scripts
- Structured Scripts
- Shared Scripts
- Data-Driven Scripts
- Keyword-driven Scripts

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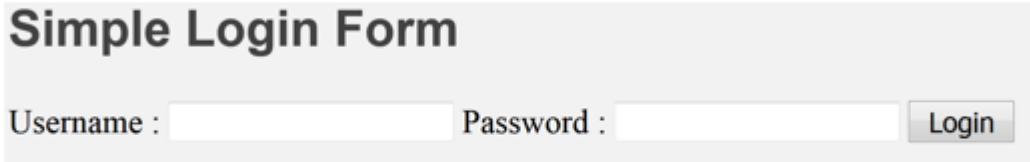
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Tools support for test execution & logging

Example: Simple Login Form

Test with different combinations of username and password



Problem: Necessary to write three scripts for three different combinations?

1. Go to login page 2. Type username “Hansen” 3. Type password“oslo123” 4. Click “Login” button	1. Go to login page 2. Type username “Olsen” 3. Type password“bergen456” 4. Click “Login” button	1. Go to login page 2. Type username “Jensen” 3. Type password“harstad789” 4. Click “Login” button

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Tools support for test execution & logging

Example: Simple Login Form

This test approach is time-consuming

Solution: Separate test script from data (username, password) → No hard-coding

One script retrieves different combinations of username and password

1. Go to page
2. Type username "file.nextUsername()"
3. Type username "file.nextPassword()"
4. Click "Login" button

Username	Password
Hansen	oslo123
Olsen	bergen456
Jensen	harstad789



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Tools support for test execution & logging

Keyword-driven scripting

Keywords symbolising actions (functionality)

"One level up" from data-driven scripting

Can write tests using keywords

"What to test, rather than how to test it"

Keyword	Script
Login	script1
CH_password	script2
Logout	script3

[script1]
1. Go to page
2. Type username "file.nextUsername()"
3. Type username "file.nextPassword()"
4. Click "Login" button

[script2]
1. Click on user avatar
2. Click "Change password"
3. Type current password
4. Type new password
5. Click "Confirm" button

[script3]
1. Click on user avatar
2. Click "Logout" button



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Tools support for test execution & logging

- Test execution tools

Coverage tools

- identifying coverage items (instrumenting the code)
- calculating the percentage of coverage items that were exercised by a suite of tests
- reporting coverage items that have not yet been exercised
- identifying test inputs to exercise items that have not yet been uncovered
- generating stubs and drivers (if part of a unit test framework).

- Test harnesses and Unit test framework tools (D)

- Test comparators

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Tools support for test execution & logging

- Test execution tools
- Coverage tools

Test harnesses and Unit test framework tools (D)

- supplying inputs to the software being tested
- receiving outputs generated by the software being tested
- executing a set of tests within the framework or using the test harness
- recording the pass/fail results of each test (framework tools)
- storing tests (framework tools)
- support for debugging (framework tools)
- coverage measurement at code level (framework tools)



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Tools support for test execution & logging

Drivers

Calls the component to be tested

In other words: A component that calls the *Tested Unit*

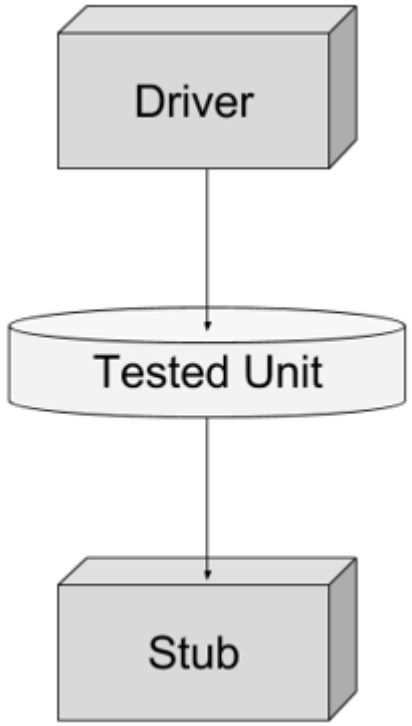
Stubs

Called *from* the software component to be tested

In other words: A component the *Tested Unit* depends on

Partial implementation

Fake values



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Tools support for test execution & logging

Test harness and unit test framework tools (D)

The two types are similar

Support tools for testing individual components or software units

Harness: Stubs and drivers → Small programs that interact with software

Unit test framework tools → Support for object-oriented software

When are these tools used?

During test execution and logging



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Tools support for test execution & logging

Characteristics of test harness and unit test framework tools

Supply inputs to the software being tested

Receive outputs generated by the software being tested

Execute a set of tests within the framework

Record pass / fail results of each test

Store tests

Coverage measurement at code level

Provide support for debugging

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Tools support for test execution & logging

- Test execution tools
- Coverage tools
- Test harnesses (D) and Unit test framework tools (D)
- **Test comparators**



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Tools support for test execution & logging

Test comparators are used when the executed test generates a lot of output.

Testing is more than providing inputs

Need to check if software produces the correct result

Compare actual outcomes to expected results

Two ways of comparing results

Dynamic comparison → Comparison done during test execution

Post-execution comparison → Comparison performed after test has finished

Software under test is no longer executing



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Tools support for test execution & logging

Test comparators are used when the executed test generates a lot of output.

Dynamic comparison

Best done by test execution tools

Useful when actual results do not match expected results in the middle of a test

Tool may be programmed to take recovery actions / go to a different set of tests

Example

Good for comparing wording of an error message

Does the pop-up message match the correct wording for that error message?

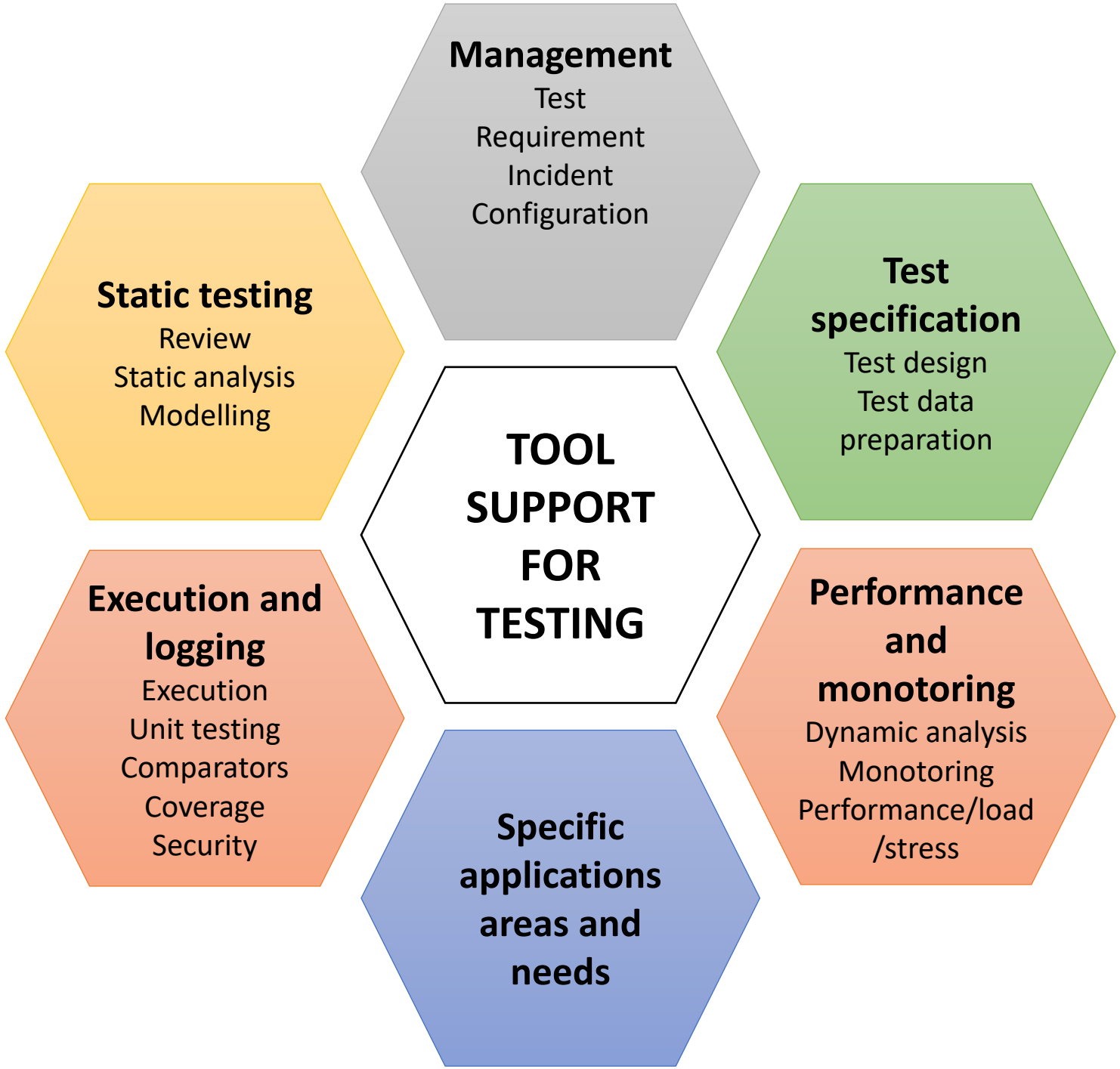


Test comparators

Tasks | Test Data Comparator

Action	Type	Test Data		
		In	Expected	Actual
objMoney = new Money(amount, curr...				
amount	int	9	[0..8]	9
currency	java.lang.String			
objMoney	junit.samples.money.Mo...			
<expected exception>	Throwable		<no exception>	<no exception>
retValue = objMoney.equals(anObject)				
anObject	java.lang.Object			
retValue	boolean			
<expected exception>	Throwable		<no exception>	<no exception>

Test Data Comparator loaded.


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Tools support for performance & monitoring

Test data for performance testing

Real data

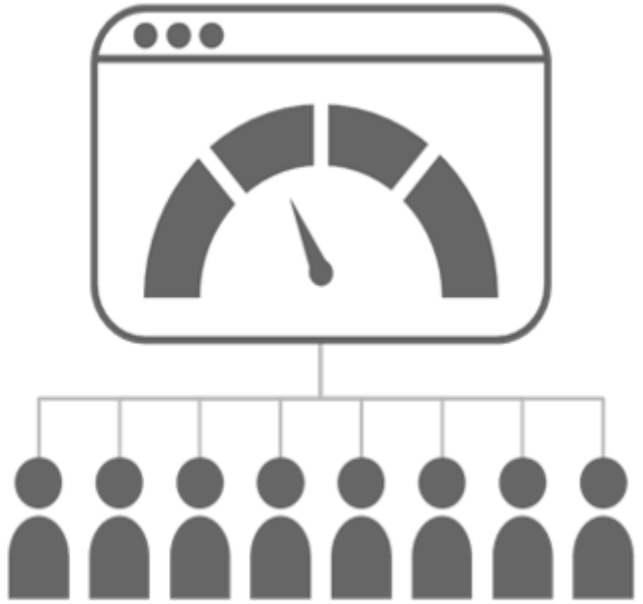
Test data obtained from users

Load

Large amounts of test data can be produced

Maintenance

Test data from the production environment



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Tools support for performance & monitoring

Tests should reflect realistic (correct) scenarios

Systems are often required to handle significant load / interactions

Inadequate / insufficient testing compromises system quality

Setting up test data → Significant effort

Extensive range or volume of data needed

Creating this data can be very resource-consuming



Test data preparation tools help us manage this effort

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Tools support for performance & monitoring

Common features of test data preparation tools

Data can be selected from an existing database

Data can be created, generated, and altered for use in tests

Construct a large number of similar records → Volume tests

When to use?

During test specification and control → Test data management is difficult

Ensure the system under test is being tested *realistically*

Useful for performance and reliability testing



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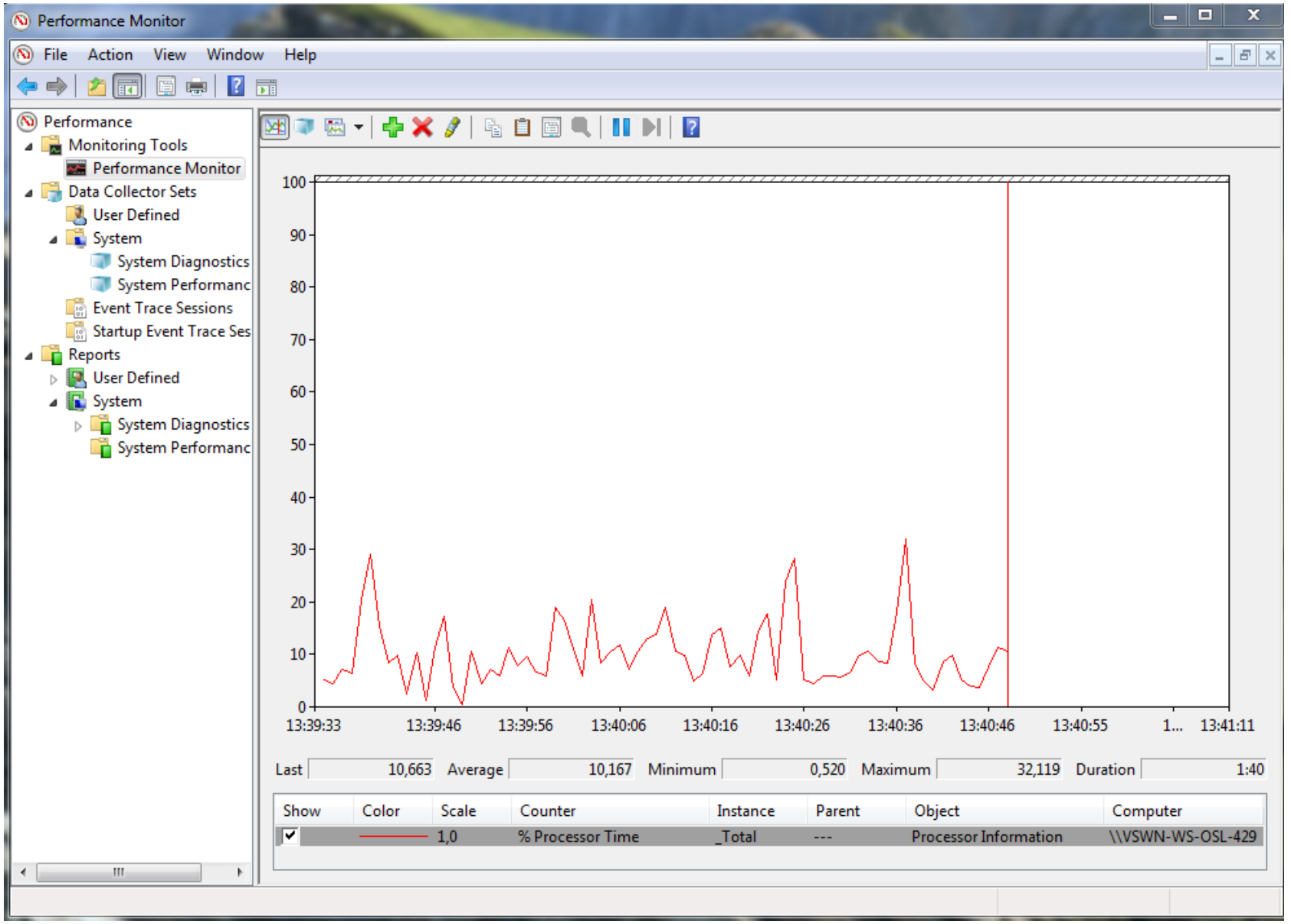
Monitoring tools

Monitoring tools identifying problems and sending an alert message to the administrator (e.g. network administrator)

- logging real-time and historical information
- finding optimal settings
- monitoring the number of users on a network
- monitoring network traffic (either in real time or covering a given length of time of operation with the analysis performed afterwards).



Monitoring tools



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Tools support for performance & monitoring

Features or characteristics of dynamic analysis tools include support for

- detecting memory leaks;
- identifying pointer arithmetic errors such as null pointers;
- identifying time dependencies



Dynamic analysis tools (D)

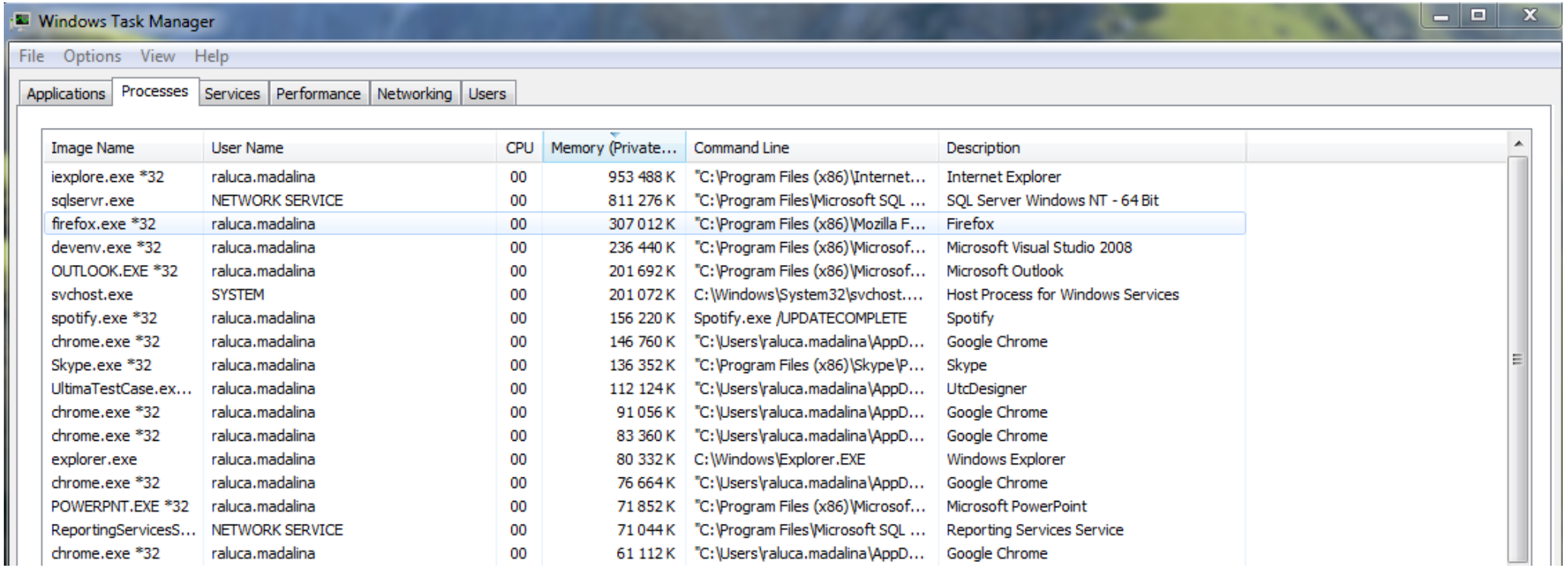
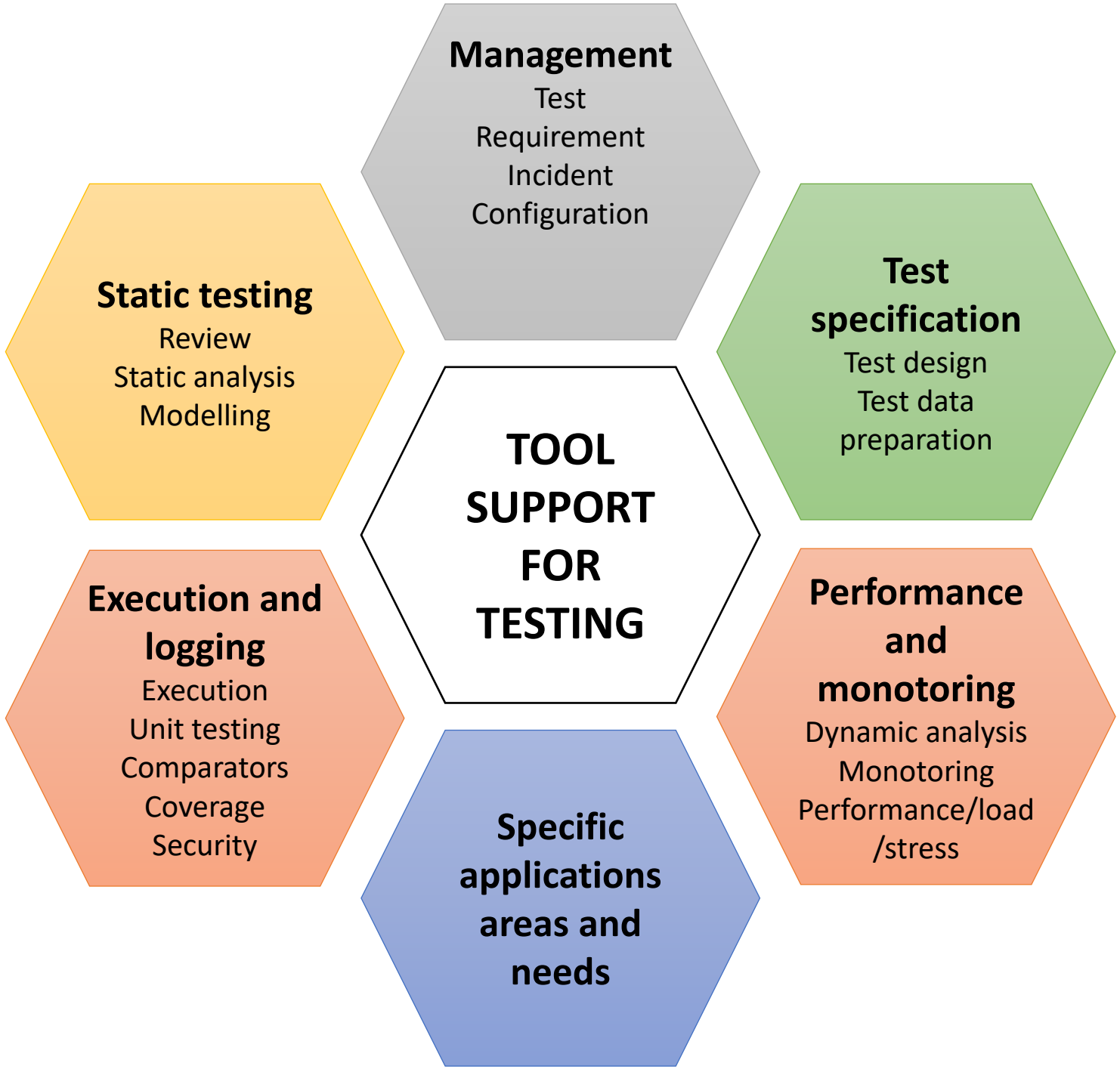


Image Name	User Name	CPU	Memory (Private...	Command Line	Description
iexplore.exe *32	raluca.madalina	00	953 488 K	"C:\Program Files (x86)\Internet...	Internet Explorer
sqlservr.exe	NETWORK SERVICE	00	811 276 K	"C:\Program Files\Microsoft SQL ...	SQL Server Windows NT - 64 Bit
firefox.exe *32	raluca.madalina	00	307 012 K	"C:\Program Files (x86)\Mozilla F...	Firefox
devenv.exe *32	raluca.madalina	00	236 440 K	"C:\Program Files (x86)\Microsof...	Microsoft Visual Studio 2008
OUTLOOK.EXE *32	raluca.madalina	00	201 692 K	"C:\Program Files (x86)\Microsof...	Microsoft Outlook
svchost.exe	SYSTEM	00	201 072 K	C:\Windows\System32\svchost....	Host Process for Windows Services
spotify.exe *32	raluca.madalina	00	156 220 K	Spotify.exe /UPDATECOMPLETE	Spotify
chrome.exe *32	raluca.madalina	00	146 760 K	"C:\Users\raluca.madalina\AppD...	Google Chrome
Skype.exe *32	raluca.madalina	00	136 352 K	"C:\Program Files (x86)\Skype\P...	Skype
UltimaTestCase.ex...	raluca.madalina	00	112 124 K	"C:\Users\raluca.madalina\AppD...	UtcDesigner
chrome.exe *32	raluca.madalina	00	91 056 K	"C:\Users\raluca.madalina\AppD...	Google Chrome
chrome.exe *32	raluca.madalina	00	83 360 K	"C:\Users\raluca.madalina\AppD...	Google Chrome
explorer.exe	raluca.madalina	00	80 332 K	C:\Windows\Explorer.EXE	Windows Explorer
chrome.exe *32	raluca.madalina	00	76 664 K	"C:\Users\raluca.madalina\AppD...	Google Chrome
POWERPNT.EXE *32	raluca.madalina	00	71 852 K	"C:\Program Files (x86)\Microsof...	Microsoft PowerPoint
ReportingServicesS...	NETWORK SERVICE	00	71 044 K	"C:\Program Files\Microsoft SQL ...	Reporting Services Service
chrome.exe *32	raluca.madalina	00	61 112 K	"C:\Users\raluca.madalina\AppD...	Google Chrome


Management
Test
Requirement
Incident
Configuration
Static testing
Review
Static analysis
Modelling
Test specification
Test design
Test data preparation
TOOL SUPPORT FOR TESTING
Execution and logging
Execution
Unit testing
Comparators
Coverage
Security
Performance and monotoring
Dynamic analysis
Monotoring
Performance/load /stress
Specific applications areas and needs

1. Types of test tools
- *1.1 Tool support for testing*
- *1.2 Test tool classification*
- *1.3 Tools for test management*
- *1.4 Tools for static testing*
- *1.5 Tools for test design and specification*
- *1.6 Tools for execution and logging*
- *1.7 Tools for performance & monitoring*
- *1.8 Tools for specific testing needs*

2. Effective use of test tools
- *2.1 Potential benefits and risks of tools*
- *2.2 Special considerations for tools*

3. Introducing a test tool into an organization

Tools support for specific application areas

There are tools specialized for use in a particular type of application

- Data quality assessment
- Data conversion and migration
- Usability testing and Accessibility testing
- Localization testing
- Security testing
- Portability testing



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Tools support for specific application areas

- There are tools specialized for use in a particular type of application.

Example:

- performance testing tools specifically for web-based applications
- dynamic analysis tools specifically for testing security aspects.

- Example of targeted areas: embedded systems.



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Tools support for specific application areas

- **Testers may use:**
  - word processor
  - spreadsheets

  as a testing tool, but they are often used to store:
  - test designs
  - test scripts
  - test data.

- Testers may also use SQL to set up and query databases containing test data.
- Tools used by developers when debugging, to help localize defects and check their fixes, are also testing tools.
- It is a good idea to look at any type of tool available to you for ways it could be used to help support any of the testing activities.

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Special considerations: Test execution tools

This type of tool often requires significant effort in order to achieve significant benefits.

- Capturing tests by recording the actions of a manual tester seems attractive, but this approach does not scale to large numbers of automated tests. This type of script may be unstable when unexpected events occur.

- **Data-driven approach**: separates out the test inputs (the data) and uses a more generic script that can read the test data and perform the same test with different data.

- **In a keyword-driven approach:** the spreadsheet contains keywords with the actions to be taken (also called action words), and test data. Testers can then define tests using the keywords.

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Special considerations: Performance testing tools

- The design of the load to be generated by the tool
- Timing aspects – probe effect
- How to interpret the information gathered.
- These tools need tester with expertise in performance testing to design the tests and interpret results.



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Special considerations: Static analysis tools

- There is a risk that the changes to make old code to conform to new standard will introduce an unexpected side-effect.
- These tools applied to source code can enforce coding standards, but if applied to existing code may generate a lot of messages.
- A gradual implementation with initial filters to exclude some messages would be an effective approach.



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Special considerations: Test management tools

- They need to interface with other tools or spreadsheets in order to produce information in the best format for the current needs of the organization.



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Introducing a test tool into an organization

The ***main considerations*** in selecting a tool for an organization include:

- Assess the organizational maturity, strengths and weaknesses
- Evaluate against clear requirements and objective criteria.
- A proof-of-concept to test the required functionality and determine whether the product meets its objectives.
- Evaluation of the vendor (including training, support and commercial aspects).
- Identification of internal requirements for coaching and mentoring in the use of the tool.



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Introducing a test tool into an organization

Introducing the selected tool into an organization starts with a ***pilot project, with the following objectives***:

- Learn more detail about the tool.
- Evaluate how the tool fits with existing processes and practices, and determine what would need to change.
- Decide on standard ways of using and maintaining the tool and the test.
- Assess whether the benefits will be achieved at reasonable cost.



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Introducing a test tool into an organization

Success factors for the deployment of the tool within an organization include:

- Roll out the tool to the rest of the organization incrementally.
- Adapt and improve processes to fit with the use of the tool.
- Provide training and coaching/mentoring for new users.
- Define usage guidelines.
- Implement a way to learn lessons from tool use.
- Monitor the tool use and benefits.

